

#1

$$P(s, t) = P_0 + s(P_1 - P_0) + t(P_2 - P_0)$$

$$(u, v) = q_0 + s(q_1 - q_0) + t(q_2 - q_0)$$

$$e_1 = q_1 - q_0, \quad e_2 = q_2 - q_0$$

$$(u, v) - q_0 = se_1 + te_2$$

$$d = se_1 + te_2 \quad \dots \quad d \equiv (u, v) - q_0$$

$$e_1 = (e_{1x}, e_{1y}), \quad e_2 = (e_{2x}, e_{2y}), \quad d = (d_x, d_y)$$

$$\begin{bmatrix} e_{1x} & e_{1y} \\ e_{2x} & e_{2y} \end{bmatrix} \begin{bmatrix} s \\ t \end{bmatrix} = \begin{bmatrix} d_x \\ d_y \end{bmatrix}, \quad \Delta = e_{1x}e_{2y} - e_{1y}e_{2x}$$

$$s = \frac{e_{2y}d_x - e_{2x}d_y}{\Delta}, \quad t = \frac{-e_{1y}d_x + e_{1x}d_y}{\Delta}$$

#2

$$P(s, t) = P_0 + s(P_1 - P_0) + t(P_2 - P_0)$$

$$P(u, v) = P_0 + s(u, v)(P_1 - P_0) + t(u, v)(P_2 - P_0)$$

$$= P_0 + \frac{e_{2y}d_x - e_{2x}d_y}{\Delta} (P_1 - P_0) + \frac{-e_{1y}d_x + e_{1x}d_y}{\Delta} (P_2 - P_0)$$

#3

$$s = \frac{e_{2y}d_x - e_{2x}d_y}{\Delta}, \quad t = \frac{-e_{1y}d_x + e_{1x}d_y}{\Delta}, \quad d_x = u - q_{0x}, \quad d_y = v - q_{0y}$$

$$\frac{\partial s}{\partial u} = \frac{e_{2y}}{\Delta}, \quad \frac{\partial s}{\partial v} = -\frac{e_{2x}}{\Delta}, \quad \frac{\partial t}{\partial u} = -\frac{e_{1y}}{\Delta}, \quad \frac{\partial t}{\partial v} = \frac{e_{1x}}{\Delta}$$

$$\frac{\partial P}{\partial u} = (P_1 - P_0) \frac{\partial s}{\partial u} + (P_2 - P_0) \frac{\partial t}{\partial u} = \frac{e_{2y}}{\Delta} (P_1 - P_0) - \frac{e_{1y}}{\Delta} (P_2 - P_0)$$

$$\frac{\partial P}{\partial v} = (P_1 - P_0) \frac{\partial s}{\partial v} + (P_2 - P_0) \frac{\partial t}{\partial v} = -\frac{e_{2x}}{\Delta} (P_1 - P_0) + \frac{e_{1x}}{\Delta} (P_2 - P_0)$$

$\frac{\partial P}{\partial u}, \frac{\partial P}{\partial v}$ 는 텍스처 공간에서 각각 변화가 표면 위 3D 위치를 어떤 방향으로 움직이는지 나타내는 접선 벡터이다.